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Nomenclature

Physical constants

ε_0	Permittivity of free space	
μ_0	Permeability of free space	
a_0	Bohr radius	
c	Speed of light in vaccuum	
e	Elementary charge	
h	Planck constant	
k_B	Boltzmann constant	
m_e	Electron mass	
N_A	Avogadro's number	
$\mathcal{R}$	Rydberg energy	

Scalars

α	Attenuation coefficient	$\mathrm{m^{-1}}$
$\dot{\alpha}$	Volume fraction rate of change	$\mathrm{s^{-1}}$
β	Thermoelectric coefficient	1
Γ	Reaction rate density	$\mathrm{m^{-3}\,s^{-1}}$
γ	Adiabatic index	1
ε_r	Relative permittivity	1
η	Refractive index	1
κ	Thermal conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
μ	Dynamic viscosity	$\mathrm{Pa\,s}$
μ_r	Relative permeability	1

ν	Frequency	Hz
ρ	Charge density	$\mathrm{C\,m^{-3}}$
ρ	Mass density	$\mathrm{kg\,m^{-3}}$
σ	Reaction cross section	$\mathrm{m^2}$
Φ	Quantum yield	1
ϕ	Scalar flux	$\mathrm{m^{-2}\,s^{-1}}$
χ	Magnetization	1
ω	Angular frequency	$\mathrm{rad\,s^{-1}}$
A	Area	$\mathrm{m^2}$
$\mathcal{E}$	Particle internal energy	J
ρE	Internal energy density	$\mathrm{J\,m^{-3}}$
E	Specific internal energy	$\mathrm{J\,kg^{-1}}$
$\mathcal{H}$	Particle enthalpy	J
ρH	Enthalpy density	$\mathrm{J\,m^{-3}}$
H	Specific enthalpy	$\mathrm{J\,kg^{-1}}$
I	Intensity	$\mathrm{W\,m^{-2}}$
M	Molar mass	$\mathrm{kg\,mol^{-1}}$
m	Mass	kg
N	Number	1
n	Number density	$\mathrm{m^{-3}}$
p	Pressure	Pa
q'''	Volumetric heat generation rate	$\mathrm{W\,m^{-3}}$
S	Reaction rate coefficient	varies
s	Distance	m
T	Temperature	K
t	Time	s
$\mathcal{U}$	Particle potential energy	J
V	Volume	$\mathrm{m^3}$

v	Speed	$\mathrm{m\,s^{-1}}$

Vectors

$\hat{\boldsymbol{\omega}}$	Direction	1
$\mathbf{a}$	Acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{B}$	Magnetic field	T
$\mathbf{E}$	Electric field	$\mathrm{V\,m^{-1}}$
$\mathbf{f}$	Force density	$\mathrm{N\,m^{-3}}$
$\mathbf{g}$	Gravitational acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{j}$	Current density	$\mathrm{A\,m^{-2}}$
$\hat{\mathbf{n}}$	Normal direction	1
$\mathbf{q}''$	Heat flux density	$\mathrm{W\,m^{-2}}$
$\mathbf{r}$	Position	m
$\mathbf{S}$	Poynting vector	$\mathrm{W\,m^{-2}}$
$\mathbf{u}$	Velocity	$\mathrm{m\,s^{-1}}$
$\rho\mathbf{u}$	Momentum density	$\mathrm{N\,s\,m^{-3}}$

Matrices

$\mathbf{K}$	Conductivity tensor	$\mathrm{W\,m^{-1}\,K^{-1}}$
$\mathbf{\Pi}$	Thermoelectric flux tensor	V
$\mathbf{T}$	Deviatoric stress tensor	Pa

Subscripts

3br	Three-body recombination
ci	Collisional ionization
ib	Inverse bremsstrahlung
pi	Photoionization
rr	Radiative recombination
γ	Photon
b	Bound electron
e	Free electron

i	Ion

Superscripts

K	Kinetic energy
M	Momentum

Chapter 1

Introduction

Chapter 2

Physics of laser-droplet interactions

Modeling the interaction between a laser beam and droplet combines the approaches of multiphase flow and magnetohydrodynamics (MHD). The former is simply due to the existence of the two immiscible liquid and gas phases, and the latter is especially necessary if the laser intensity is strong enough to ionize. In this paper, the term gas phase includes both neutral and charged particles, as both are governed by the ideal gas equation of state.

In order to accurately capture the behavior of the liquid-gas-plasma, a four-fluid model is used to describe the constituent fluids – namely, liquid, gas, electrons and ions – that interact with one another. The charged particles also exhibit electromagnetic effects. The governing equations are therefore derived from the three conservation laws and Maxwell's equations.

Chemically speaking, the laser can depart enough energies to not only remove bound electrons from their orbits but also break molecular bonds and alter the chemical composition of the fluid. In fact, free radicals and light molecular ions are frequently encountered in plasmas if the laser is not strong enough to completely disintegrate molecules into monoatomic ions. However, for simplicity, all ions are assumed to have resulted from neutral molecules each losing a single electron, and thus they have a charge of +1 and the same molecular weight as the parent neutral particle.

2.1 Fluid description of plasmas

2.1.1 Conservation equations

The multiphase flow model is based on the Baer-Nunziato model of a liquid-vapor system not in equilibrium. The model allows for distinct velocities, temperatures, and pressures between the phases and employs relaxation terms to drive those quantities into equilibrium. To extend this model to include the plasma state, it is first assumed that charged particles will only occupy the gas phase and interact solely with the neutral gas particles. The gas phase then acts the intermediate phase between liquid and plasma, and its flow equations have to include interactions (phase change and relaxations) with both. The relaxation terms between fluids j and k in the volume fraction, momentum, and energy equations are denoted $\dot{\alpha}_{jk}$, $\mathbf{f}_{jk}$, q'''_{jk}, respectively.

There are four fluids in this system, and the subscripts (and sometimes superscript) to identify them are l for liquid, g for neutral gas, e for electrons, and i for ions. One exception exists for particle mass m (a single particle does not exhibit state behavior), in which case m_e stands for electron mass and m stands for the neutral particle or ion mass. For laser interaction term, γ is also used to indicate photons (see Section 2.6). This convention is used for the rest of the document.

The conservation equations for the liquid phase includes the mass, momentum, and energy equations on top of an additional equation for the liquid volume fraction, α_l.

$$\frac{\partial \alpha_l}{\partial t} + \nabla \cdot \alpha_l \mathbf{u}_l = \dot{\alpha}_{gl} \tag{2.1a}$$

$$\frac{\partial \alpha_l \rho_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l = 0 \tag{2.1b}$$

$$\frac{\partial \alpha_l \rho_l \mathbf{u}_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l \otimes \mathbf{u}_l = -\nabla p_l + \nabla \cdot \mathbf{T}_l + \mathbf{f}_{gl} \tag{2.1c}$$

$$\frac{\partial \alpha_l \rho_l E_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l H_l = -\nabla \cdot \mathbf{q}_l'' + \nabla \cdot [\mathbf{T}_l \cdot \mathbf{u}_l] + q_{gl}'''. \tag{2.1d}$$

The neutral gas equations are written similarly, but the gas volume fraction α_g can be computed algebraically as $\alpha_g = 1 - \alpha_l$:

$$\frac{\partial \alpha_g \rho_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g = 0 \tag{2.2a}$$

$$\frac{\partial \alpha_g \rho_g \mathbf{u}_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g \otimes \mathbf{u}_g = -\nabla p_g + \nabla \cdot \mathbf{T}_g + \sum_j^{l,e,i} \mathbf{f}_{jg} \tag{2.2b}$$

$$\frac{\partial \alpha_g \rho_g E_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g H_g = -\nabla \cdot \mathbf{q}_g'' + \nabla \cdot [\mathbf{T}_g \cdot \mathbf{u}_g] + \sum_j^{l,e,i} q_{jg}'''. \tag{2.2c}$$

For charge particles, formulae relating physical quantities are more conveniently expressed using number rather than mass density ρ, as many electromagnetic quantities and plasma properties are calculated using number density n and charge density en. As a result, momentum density is written as $m_j n_j \mathbf{u}_j$, and energy density as $n_j \mathcal{E}_j$. The electron conservation equations are

$$\frac{\partial \alpha_g n_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e = 0 \tag{2.3a}$$

$$m_e \left[\frac{\partial \alpha_g n_e \mathbf{u}_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e \otimes \mathbf{u}_e \right] = -\nabla p_e + \nabla \cdot \mathbf{T}_e + \sum_j^{g,i} \mathbf{f}_{je} \tag{2.3b}$$

$$\frac{\partial \alpha_g n_e \mathcal{E}_e}{\partial t} + \nabla \cdot \alpha_g \rho_e \mathbf{u}_e \mathcal{H}_e = -\nabla \cdot \mathbf{q}_e'' + \nabla \cdot [\mathbf{T}_e \cdot \mathbf{u}_e] + \sum_j^{g,i} q_{je}'''. \tag{2.3c}$$

Finally, the ion conservation equations are

$$\frac{\partial \alpha_g n_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i = 0 \tag{2.4a}$$

$$m \left[\frac{\partial \alpha_g n_i \mathbf{u}_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i \otimes \mathbf{u}_i \right] = -\nabla p_i + \nabla \cdot \mathbf{T}_i + \sum_j^{g,e} \mathbf{f}_{ji} \tag{2.4b}$$

$$\frac{\partial \alpha_g n_i \mathcal{E}_i}{\partial t} + \nabla \cdot \alpha_g \rho_i \mathbf{u}_i \mathcal{H}_i = -\nabla \cdot \mathbf{q}_i'' + \nabla \cdot [\mathbf{T}_i \cdot \mathbf{u}_i] + \sum_j^{g,e} q_{ji}'''. \tag{2.4c}$$

2.1.2 Relaxation terms

This section details the relaxation processes between the liquid and gas phases. For details about relaxations of charge particles, see Section 2.3.1.

2.1.3 Reaction terms

Reaction and phase change are used interchangeably in this document, since the same chemical species of different states are treated as different fluids. A reaction rate density Γ_r^j describes how many reactions occurs per unit volume per unit time. The subscript r indicates the reaction type and the superscript j indicates the phase of interest that is involved in said reaction. The term is negative for reactants and positive for products. In a binary reaction – for example, boiling – the sum of the two Γ terms is 0; therefore, $\Gamma^l_{\text{boil}} = -\Gamma^g_{\text{boil}}$. Ternary reactions such as ionization and recombination will see an imbalance of particles following a reaction. Section **??** describes the most commonly found ionization reactions encountered in the system.

Reaction also transfer a portion of the reactant's momentum and energy into the product pool, and the particle momentum $m_j\mathbf{u}_j$ and energy $\mathcal{E}_j$ being transferred are those of the reactants. When expressing this transfer under a summation operator, e.g., $\sum_r \Gamma_r^j \mathcal{E}_r$, Γ_r^j indicates all reactions r that involve particle j, and $\mathcal{E}_r$ means the energy of the reactant in each reaction. Note that this only applies readily for heavy particles; electrons typically involve reaction-specific transfer terms that are described in Section 2.5.

If it is convenient to express reaction rate density in mass-based units, the conversion factor is the mass of the particle of interest, m.

2.1.4 Transport terms

The diffusive terms in the momentum and energy equations are $\nabla \cdot \mathbf{T}$ and $\nabla \cdot \mathbf{q}''$, respectively. For neutral particles, the deviatoric stress tensor $\mathbf{T}$ is

$$\mathbf{T} = \mu \left[\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right], \tag{2.5}$$

and the isotropic heat flux vector $\mathbf{q}''$ is

$$\mathbf{q}'' = -\kappa \nabla T. \tag{2.6}$$

The viscosity μ and thermal conductivity κ are given by a material property library. Transport properties of charged particles are described in Section 2.4.

2.1.5 Closure relations

For each set of conservation equations, a closer relation is required to close the system. It usually takes form of an equation of state and relates a conserved variable to a set of primitive variables. Species in the gas phase, neutral or charged, follow an ideal gas assumption.

2.2 Electromagnetic effects

2.2.1 Maxwell's equations

Hydrocarbon materials such that one used in this project can be reasonably assumed to be isotropic and homogeneous. This simplifies the Maxwell's equations in matter because it allows two vectors that are difficult to compute, namely the polarization vector $\mathbf{P}$ and the magnetization vector $\mathbf{M}$, to be parallel to the electric and magnetic fields. Moreover, $\mu_r \approx 1$ is assumed since hydrocarbon are non-magnetic materials. This leads to the following form of Maxwell's equations in matter:

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\epsilon_r \epsilon_0} \tag{2.7a}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2.7b}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{2.7c}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}_f + \mu_0 \epsilon_r \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{2.7d}$$

In plasma, the free charge density ρ_f and current density $\mathbf{j}_f$ can be easily computed using electron and ion information:

$$\rho_f \equiv e(n_i - n_e) \tag{2.8a}$$

$$\mathbf{j}_f \equiv e(n_i \mathbf{u}_i - n_e \mathbf{u}_e) \tag{2.8b}$$

Maxwell's Equations (Equation 2.7) are over-determined; there are 8 scalar equations that only describes 6 scalar variables (the components of $\mathbf{E}$ and $\mathbf{B}$). However, it can be shown that taking the divergence of Equations 2.7c and 2.7d will lead to Equations 2.7b and 2.7a, respectively. Therefore, only Equations 2.7c and 2.7d are necessary to model the $\mathbf{E}$ and $\mathbf{B}$ fields.

2.2.2 Lorentz force

The Lorentz force acts upon a distribution of charged particles has the form

$$\mathbf{f}_j = \pm e n_j \left(\mathbf{E} + \mathbf{u}_j \times \mathbf{B} \right). \tag{2.9}$$

Since electrons are extremely lightweight, its acceleration due to the Lorentz force easily dominates any diffusive or gravitational effects, so they are safe to neglect. The case for ions is a little bit more nuanced; however, the Lorentz force is still expected to be the main momentum contributor.

Electromagnetic waves themselves carry energy, and therefore capable of doing work on a charged system. According to Equation 2.9, the magnetic field does no work on charged particles as its force component is always perpendicular to the charge velocity. The rate of work done by the electric field is then

$$W_j = \mathbf{f}_j \cdot \mathbf{u}_j = \pm e n_j \mathbf{u}_j \cdot \mathbf{E}, \tag{2.10}$$

and should be equal to the rate of energy gained by the particles. This accounts for the change in kinetic energy of a charged particles under the influence of the electric fields.

2.3 Collisions in plasmas

Species in a plasma exchange their mass, momentum, and energy primarily through local collisions. The collisions described in this section are assumed to be entirely binary; i.e. involving two particles at a time. There are two interactions that can follow a local collision of unlike particles, namely reaction and relaxation. Reactions convert a neutral particles into ions or vice versa, and relaxations drive the system towards equilibrium in absence of external sources.

2.3.1 Relaxations

A collision between a charged particle j and a neutral particle results in a force roughly equal to

$$\mathbf{f}_{jn} = \frac{m_j n_j (\mathbf{u}_j - \mathbf{u})}{\tau_{jn}}. \tag{2.11}$$

Here the double subscript jn refers to a transfer from a particle j to a neutral particle. By Newton's third law of motion, $\mathbf{f}_{jn} = -\mathbf{f}_{nj}$. The mean time between collisions τ_{jn} is approximately constant and is the reciprocal of the momentum-transfer collision frequency

$$\nu_{jn}^M \equiv \frac{1}{\tau_{jn}} = n\sigma_{jn}^M \|\mathbf{u}_j - \mathbf{u}\|, \tag{2.12}$$

with σ_{jn}^M denotes the equivalent cross section of momentum transfer between j and n. It is typically a function of charged particle speed; the average value over all velocities in a Maxwellian distribution is typically used in Equation 2.12.

Collisions between two charged particles are Coulomb collisions mediated by the electric field. Like-particle (electron-electron or ion-ion) collisions are often neglected because they preserve the momentum and energy between themselves, and they contribute very little to diffusion. This is in contrast with electron-ion collisions, where there exists a net transfer of momentum (and energy) between the two particles. In a similar formulation as Equation 2.11, the force exerted between an ion and an electron during their collision is

$$\mathbf{f}_{ei} = m_e n_e (\mathbf{u}_i - \mathbf{u}_e)\nu_{ei}^M, \tag{2.13}$$

and the electron-ion momentum-transfer collision frequency ν_{ei}^M is

$$\nu_{ei}^M = C\frac{n_i Z^2 e^4}{(4\pi\varepsilon_0)^2\sqrt{m_e (k_B T_e)^3}} \ln\Lambda_e, \tag{2.14}$$

where C is a prefactor on the order of $\mathcal{O}(1)$ that differs depending on the text. Chen gives $C = \pi$, while the FLASH code uses $C = \frac{4}{3}\sqrt{2\pi}$ [2][3]. The latter will be used in this model. The Coulomb logarithm $\ln\Lambda$ is a correction factor that accounts for the cumulative effect of many small-angle deflections in a collision and is given by

$$\ln\Lambda_e \equiv \ln\left[\frac{4\pi(\varepsilon_0 k_B T_e)^{3/2}}{Ze^3\sqrt{n_e}}\right]. \tag{2.15}$$

In both Equations 2.14 and 2.15, the average ionization number Z can be assumed to be 1.

For all types of collisions, the energy loss due to relaxation term can be expressed as a heat source that flows from particle j to j'

$$q_{jj'}''' = -q_{j'j}''' = \frac{3}{2} n_j \nu_{jj'}^K k_B (T_j - T_{j'}). \tag{2.16}$$

The energy-transfer collision frequency ν^K is related to its momentum counterpart ν^M by

$$\frac{\nu_{jj'}^K}{\nu_{jj'}^M} = \frac{2m_j}{m_j + m_{j'}}. \tag{2.17}$$

The index j in Equation 2.17 refers to the lighter particle out of the pair, so it is always the electron if it is involved in a collision, and the ion in a ion-neutral collision. In the former case, $\nu_{ej'}^K \ll \nu_{ej'}^M$, which means energy is relaxed at a much slower rate than momentum is. Physically, upon colliding with a heavy particle, an electron is frequently scattered before a significant portion of its kinetic energy is lost. In an ion-neutral collision, due to the near-identical mass of the two particles, energy and momentum are relaxed on the same timescale – i.e., $\nu_{in}^K = \nu_{in}^M$.

2.4 Charged particle transport

Transport of neutral particles are caused by the "random walk" movement of the particles, and because there is no inherent preference for a particular direction, the transported quantity – being momentum or energy – is eventually equilibrated. This is also true in the case of charged particles in the *unmagnetized* limits; that is when $\|\mathbf{B}\| = 0$. The existence of a magnetic field gives rise to anisotropy in heat conduction, since the particles are forced to spiral along the magnetic field lines, so the rate of transport is the highest in the direction parallel to the magnetic field lines, i.e. $\hat{\mathbf{b}} = \frac{\hat{\mathbf{B}}}{\|\mathbf{B}\|}$. On the other hand, motion across the field lines (the perpendicular direction) are strongly suppressed, so there is little transport in that direction. The other direction, which is perpendicular to both $\hat{\mathbf{b}}$ and the gradient of the transported quantity and termed *cross* direction in this document, is the result of the Righi–Leduc effect.

Transport processes in plasmas are pioneered by Braginskii in his 1965 article [1], where the transport properties – namely viscosity and thermal conductivity – are given as function of magnetization χ (also called the Hall parameter). It is defined as

$$\chi_j \equiv \frac{\omega_{\mathrm{ce},j}}{\nu_{ji}},\ j = e, i \tag{2.18}$$

and the electron or ion gyrofrequency is

$$\omega_{\mathrm{ce},j} \equiv \frac{eB}{m_j}. \tag{2.19}$$

The electron-ion collision frequency is given in Equation 2.14, and the ion-ion collision frequency is

$$\nu_{ii} = \frac{4}{3}\sqrt{\pi}\frac{n_i Z^4 e^4}{(4\pi\varepsilon_0)^2\sqrt{m(k_B T_i)^3}}\ln\Lambda_i. \tag{2.20}$$

2.4.1 Heat flux

In the presence of a magnetic field, the ion heat flux is as followed:

$$\mathbf{q}_i'' = -\kappa_\parallel^i \nabla_\parallel T_i - \kappa_\perp^i \nabla_\perp T_i + \kappa_\times^i \nabla_\times T_i \tag{2.21}$$

The three components of ∇T_i,

$$\nabla_\parallel T_i = (\hat{\mathbf{b}} \cdot \nabla T_i)\hat{\mathbf{b}} \tag{2.22a}$$

$$\nabla_\perp T_i = \nabla T_e - \nabla_\parallel T_i \tag{2.22b}$$

$$\nabla_\times T_i = \hat{\mathbf{b}} \times \nabla T_i, \tag{2.22c}$$

represent, in the respective order, the parallel, perpendicular, and cross components of ∇T_i with respect to the magnetic field direction $\hat{\mathbf{b}}$.

Using the definitions in Equation 2.22, Equation 2.21 can be written directly in terms of ∇T_i:

$$\mathbf{q}_i'' = -\kappa_\parallel^i(\hat{\mathbf{b}} \cdot \nabla T_i)\hat{\mathbf{b}} - \kappa_\perp^i(\nabla T_i - (\hat{\mathbf{b}} \cdot \nabla T_i)\hat{\mathbf{b}}) + \kappa_\times^i \hat{\mathbf{b}} \times \nabla T_i \tag{2.23a}$$

$$= -\underbrace{\left(\kappa_\perp^i \mathbf{I} + (\kappa_\parallel^i - \kappa_\perp^i)\hat{\mathbf{b}}\hat{\mathbf{b}}^T - \kappa_\times^i \mathbf{B}_\times\right)}_{\mathbf{K}_i} \nabla T_i, \tag{2.23b}$$

where $\mathbf{B}_\times$ is the antisymmetric matrix

$$\mathbf{B}_\times = \begin{bmatrix} 0 & -b_z & b_y \\ b_z & 0 & -b_x \\ -b_y & b_x & 0 \end{bmatrix}, \tag{2.24}$$

that satisfies $\mathbf{B}_\times \mathbf{v} = \hat{\mathbf{b}} \times \mathbf{v}$ for any vector $\mathbf{v}$.

The ion thermal conductivities are given as rational functions of χ_i in Equation 4.40 of Braginskii (1965) [1]:

$$\kappa_\parallel^i = \hat{\kappa}_i \frac{{\gamma_0'}^i}{\delta_0^i}, \tag{2.25a}$$

$$\kappa_\perp^i = \hat{\kappa}_i \frac{{\gamma_1'}^i \chi_i^2 + \gamma_0'}{\chi_i^4 + \delta_1^i \chi_e^2 + \delta_0^i}, \tag{2.25b}$$

$$\kappa_\times^i = \hat{\kappa}_i \frac{\chi_i(\gamma_1'' \chi_i^2 + \gamma_0'')}{\chi_i^4 + \delta_1^i \chi_e^2 + \delta_0^i}, \tag{2.25c}$$

with the prefractor $\hat{\kappa}_i$ defined as

$$\hat{\kappa}_i \equiv \frac{n_i k_B^2 T_i}{m_i \nu_{ii}}. \tag{2.25d}$$

In the unmagnetized limit ($\chi_i \to 0$), the parallel and perpendicular κs are identical while the cross term $\kappa_\times^i$ vanishes. The isotropic flux is recovered:

$$\mathbf{q}_i'' = -\kappa_i \nabla T_i \tag{2.26}$$

The electron heat flux has a conductive component similar to that of ion, but also contains a thermoelectric term that is expressed in terms of the current density $\mathbf{j}$:

$$\begin{aligned} \mathbf{q}_e'' &= -\kappa_\parallel^e \nabla_\parallel T_e - \kappa_\perp^e \nabla_\perp T_e - \kappa_\times^e \nabla_\times T_e - \frac{k_B T_e}{e}\left(\beta_\parallel \mathbf{j}_\parallel + \beta_\perp \mathbf{j}_\perp + \beta_\times \mathbf{j}_\times\right) && (2.27a) \\ &= -\mathbf{K}_e \nabla T_e - \mathbf{\Pi}\mathbf{j}, && (2.27b) \end{aligned}$$

where the conductivity and thermoelectric flux tensors are

$$\mathbf{K}_e \equiv \kappa_\perp^e \mathbf{I} + (\kappa_\parallel^e - \kappa_\perp^e)\hat{\mathbf{b}}\hat{\mathbf{b}}^T + \kappa_\times^e \mathbf{B}_\times \tag{2.27c}$$

$$\mathbf{\Pi} \equiv \frac{k_B T_e}{e}\left[\beta_\perp \mathbf{I} + (\beta_\parallel - \beta_\perp)\hat{\mathbf{b}}\hat{\mathbf{b}}^T + \beta_\times \mathbf{B}_\times\right] \tag{2.27d}$$

The κ^e's and β's are given in Table 2 Braginskii (1965), since in general they have a dependency on Z, though in this paper $Z = 1$ is assumed [1]. Another source of electron transport properties is Epperlein and Haines (1986), and they gives a more accurate expression of perpendicular and cross properties in the moderate χ_e domain ($0.3 \leq \chi_e \leq 30$) [4]. Since this article does not contain the equivalent formulae for ion properties, it is not used in favor of the Braginskii expressions, but is cited here as a reference. The electron conductivities are

$$\kappa_\parallel^e = \hat{\kappa}_e \frac{{\gamma'}_0^e}{\delta_0^e}, \tag{2.28a}$$

$$\kappa_\perp^e = \hat{\kappa}_e \frac{{\gamma'}_1^e \chi_e^2 + {\gamma'}_0}{\chi_e^4 + \delta_1^e \chi_e^2 + \delta_0^e}, \tag{2.28b}$$

$$\kappa_\times^e = \hat{\kappa}_e \frac{\chi_e({\gamma''}_1^e \chi_e^2 + {\gamma''}_0^e)}{\chi_e^4 + \delta_1^e \chi_e^2 + \delta_0^e}, \tag{2.28c}$$

with the prefractor $\hat{\kappa}_e$ defined as

$$\hat{\kappa}_e \equiv \frac{n_e k_B^2 T_e}{m_e \nu_{ei}^M}. \tag{2.28d}$$

And the thermoelectric coefficients are

$$\beta_{\parallel} = \frac{\beta_0'}{\delta_0}, \tag{2.29a}$$

$$\beta_{\perp} = \frac{\beta_1' \chi_e^2 + \beta_0'}{\chi_e^4 + \delta_1^e \chi_e^2 + \delta_0^e}, \tag{2.29b}$$

$$\beta_{\times} = \frac{\chi_e(\beta_1'' \chi_e^2 + \beta_0'')}{\chi_e^4 + \delta_1^e \chi_e^2 + \delta_0^e}. \tag{2.29c}$$

In the unmagnetized limit ($\chi_e \to 0$), the electron heat flux is

$$\mathbf{q}_e'' = -\kappa_e \nabla T_e - \frac{\beta k_B T_e}{e}\mathbf{j}. \tag{2.30}$$

2.4.2 Momentum flux

Braginskii gives the total stress tensor as the sum of five component tensors [1]:

$$\mathbf{T} = \sum_{n=0}^{4} \mathbf{T}_n \tag{2.31a}$$

$$\mathbf{T}_0 = 3\mu_0 \left(\hat{\mathbf{b}}\hat{\mathbf{b}}^T - \frac{1}{3}\mathbf{I}\right)(\hat{\mathbf{b}} \cdot \mathbf{W}\hat{\mathbf{b}}) \tag{2.31b}$$

$$\mathbf{T}_1 = \mu_1 \left[\mathbf{PWP} + \frac{1}{2}\mathbf{P}(\hat{\mathbf{b}} \cdot \mathbf{W}\hat{\mathbf{b}})\right] \tag{2.31c}$$

$$\mathbf{T}_2 = \mu_2 \left[\mathbf{PW}\hat{\mathbf{b}}\hat{\mathbf{b}}^T + \hat{\mathbf{b}}\hat{\mathbf{b}}^T\mathbf{WP}\right] \tag{2.31d}$$

$$\mathbf{T}_3 = -\frac{\mu_3}{2}\left[\mathbf{B}_\times \mathbf{WP} - \mathbf{PWB}_\times\right] \tag{2.31e}$$

$$\mathbf{T}_4 = -\mu_4 \left[\mathbf{B}_\times \mathbf{W}\hat{\mathbf{b}}\hat{\mathbf{b}}^T - \hat{\mathbf{b}}\hat{\mathbf{b}}^T\mathbf{WB}_\times\right], \tag{2.31f}$$

where the projection matrix $\mathbf{P}$ is

$$\mathbf{P} \equiv \mathbf{I} - \hat{\mathbf{b}}\hat{\mathbf{b}}^T, \tag{2.31g}$$

and the rate-of-strain tensor $\mathbf{W}$ is

$$\mathbf{W} \equiv \left[\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right] - \frac{2}{3}\mathbf{I}\nabla \cdot \mathbf{u} \tag{2.31h}$$

The viscosities are rational functions of χ, and their polynomial coefficients are given in Equations 4.44 through

4.45 of Braginskii [1]:

$$\mu_0^j = \hat{\mu}_j \frac{\alpha_0^{j'}}{a_0^j} \tag{2.32a}$$

$$\mu_2^j = \hat{\mu}_j \frac{\alpha_2^{j'} \chi_j^2 + \alpha_0^{j'}}{\chi_j^4 + a_2^j \chi_j^2 + a_0^j} \tag{2.32b}$$

$$\mu_4^j = \hat{\mu}_j \frac{\chi_j (\chi_j^2 + \alpha_0^{j''})}{\chi_j^4 + a_2^j \chi_j^2 + a_0^j} (\delta_{ji} - \delta_{je}) \tag{2.32c}$$

$$\mu_1^j = \mu_2^j (2\chi_j) \tag{2.32d}$$

$$\mu_3^j = \mu_4^j (2\chi_j), \tag{2.32e}$$

with the leading prefractor $\hat{\mu}_j$ defined as

$$\hat{\mu}_j \equiv \frac{n_j k_B T_j}{\nu_{ji}^M}. \tag{2.32f}$$

As a note, the term $(\delta_{ji} - \delta_{je})$ evaluates to 1 for ions and -1 for electrons.

In the unmagnetized limit ($\chi_e \to 0$), the gyroviscous terms $\mathbf{T}_3$ and $\mathbf{T}_4$ vanish, while $\mu_0 = \mu_1 = \mu_2 = \mu$, and the stress tensor becomes

$$\mathbf{T} = \mu \mathbf{W}, \tag{2.33}$$

which is identical to the conventional deviatoric stress tensor.

2.5 Reactions in plasmas

The term reaction is used in this section to denote the collisions that result in the conversion between a neutral atom and an electron-ion pair. If the neutral atom is consumed, the reaction is ionization, and the inverse is called recombination. Reactions do not just alter the amount of each species in a plasma; it also transfer a portion of the reactant's momentum and energy directly into the product, assuming no enthalpy of reaction. These transfers are direct inverses in the case of ions and neutrals as followed

$$\frac{\partial n_i}{\partial t} = -\frac{\partial n_g}{\partial t} = \sum_r \Gamma_r^i \tag{2.34a}$$

$$\frac{\partial n_i \mathbf{u}_i}{\partial t} = -\frac{1}{m_i} \frac{\partial \rho_g \mathbf{u}_g}{\partial t} = \sum_r \Gamma_r^i \mathbf{u}_r \tag{2.34b}$$

$$\frac{\partial n_i \mathcal{E}_i}{\partial t} = -\frac{\partial \rho_g E_g}{\partial t} = \sum_r \Gamma_r^i \mathcal{E}_r, \tag{2.34c}$$

where the subscript r on $\mathbf{u}_r$ and $\mathcal{E}_r$ indicates that the respective variable belongs to the reactant species.

For electrons, the momentum transfer can safely be approximated as that of ions scaled by the mass ratio of the two particles. The energy source term is more complex and cannot be approximated as such; it is typically denoted as q_r^e for each reaction r. When an electron is liberated during ionization, it receives the energy of the source particle minus the binding energy required for removing the electron from its bound state. On the other hand, when an electron is captured by an ion, its energy tends to be transfered to a third particle. Each reaction will detail what its respective q_r^e should be.

$$\frac{\partial n_e}{\partial t} = \sum_r \Gamma_r^e \tag{2.35a}$$

$$\frac{\partial n_e \mathbf{u}_e}{\partial t} = \sum_r \left(\frac{m_e}{m_r}\right) \Gamma_r^e \mathbf{u}_r \tag{2.35b}$$

$$\frac{\partial n_e \mathcal{E}_e}{\partial t} = \sum_r q_r^e \tag{2.35c}$$

2.5.1 Collisional ionization

When a free electron collides with a bound electron and departs sufficient energy on it, the latter can be removed from its orbit. Known as collisional ionization, this process is responsible for rapid increase of the free electron population. Not all free electrons can cause collisional ionization; it must have a kinetic energy $\mathcal{E}_e$ that exceeds the binding energy $\mathcal{E}_b$ of the bound electron. Under the fluid model of plasma, electrons in any control volume is modeled as having a single particle-averaged energy. In this section, this averaged energy is denoted $\frac{3}{2}k_B T_e$ in this section as opposed to $\mathcal{E}_e$, which instead represents the energy of an individual electron. Collisional ionization still occurs in a volume where $\frac{3}{2}k_B T_e < \mathcal{E}_b$ since there exists a fraction of electron with a sufficient $\mathcal{E}_e$.

Assuming a Maxwellian distribution of velocity, the probability density function (PDF) of electron speed $f(u_e)$ is [5]

$$f(u_e) = \left(\frac{m_e}{2\pi k_B T_e}\right)^{3/2} 4\pi u_e^2 \exp\frac{-\frac{1}{2}m_e u_e^2}{k_B T_e}. \tag{2.36}$$

To convert Equation 2.36 into a PDF in kinetic energy, the following identity is employed

$$f(\mathcal{E}_e)\left|d\mathcal{E}_e\right| = f(u_e)\left|du_e\right|. \tag{2.37}$$

which requires the derivative

$$\frac{d\mathcal{E}_e}{du_e} = m_e u_e \tag{2.38a}$$

$$\frac{du_e}{d\mathcal{E}_e} = \frac{1}{m_e u_e} = \frac{1}{\sqrt{2m_e\mathcal{E}_e}} \tag{2.38b}$$

The energy-PDF is therefore

$$f(\mathcal{E}_e) = f(u_e)\left|\frac{du_e}{d\mathcal{E}_e}\right| \tag{2.39a}$$

$$= \frac{2}{\sqrt{\pi}}\sqrt{\frac{\mathcal{E}_e}{(k_B T_e)^3}}\exp\frac{-\mathcal{E}_e}{k_B T_e} \tag{2.39b}$$

The electron-impart ionization cross section is calculated using the Binary-Encounter-Bethe (BEB) model [7]:

$$\sigma_{\mathrm{BEB}} = \frac{4\pi a_0^2 r^2}{t+u+1}\left[\frac{\ln t}{2}\left(1-\frac{1}{t^2}\right)+1-\frac{1}{t}-\frac{\ln t}{t+1}\right], \tag{2.40a}$$

where the following dimensionless energies are used

$$r \equiv \frac{\mathcal{R}}{\mathcal{E}_b} \tag{2.40b}$$

$$t \equiv \frac{\mathcal{E}_e}{\mathcal{E}_b} \tag{2.40c}$$

$$u \equiv \frac{\mathcal{U}}{\mathcal{E}_b} \tag{2.40d}$$

The reaction rate density is

$$\Gamma_{\text{ci}} = S_{\text{ci}} n_e n_g \tag{2.41a}$$

$$S_{\text{ci}} \equiv \int_{\mathcal{E}_b}^{\infty} \sigma_{\text{BEB}} v f(\mathcal{E}_e) d\mathcal{E}_e. \tag{2.41b}$$

The rate constant S_{ci} has units of volumetric flow rate (m^3/s).

Upon ionization, the high-energy tail is depleted to create colder electrons, leading to a loss of energy. The rate of energy loss by the electron is

$$q^e_{\text{ci}} = -\Gamma_{\text{ci}} \mathcal{E}_b. \tag{2.42}$$

2.5.2 Recombination

When an electron and an ion combines to form a neutral atom, a third particle must be ejected in order to preserve momentum and energy. If that particle is a photon, that it is a *radiative recombination* reaction, which is the inverse of photoionization. According to Milne relation,

$$\sigma_{\text{rr}} = \frac{g_n}{g_i} \left(\frac{h\nu}{m_e c u_e} \right)^2 \sigma^{(1)}_{\text{pi}}, \tag{2.43}$$

and the energy of the ejected photon must be

$$h\nu = \mathcal{E}_b + \mathcal{E}_e. \tag{2.44}$$

The (single-)photoionization cross section $\sigma^{(1)}_{\text{pi}}$ is given in Equation 2.62.

Again, the rate coefficient needs to the distribution of electron energy, not just the average energy. Fortunately, recombination does not have an energy floor, so the lower bound of the energy integral is 0. The reaction rate density is

$$\Gamma_{\text{rr}} = S_{\text{rr}} n_e n_i \tag{2.45a}$$

$$S_{\text{rr}} \equiv \int_0^{\infty} \sigma_{\text{rr}} v f(\mathcal{E}_e) d\mathcal{E}_e. \tag{2.45b}$$

And the rate of electron energy loss is

$$q^e_{\text{rr}} = -K_{\text{rr}} n_e n_i \tag{2.46a}$$

$$K_{\text{rr}} \equiv \int_0^{\infty} \mathcal{E}_e \sigma_{\text{rr}} v f(\mathcal{E}_e) d\mathcal{E}_e. \tag{2.46b}$$

The integrals in Equations 2.45 and 2.46 assume a Maxwellian distribution of electron energy but the ions are monoenergetic, which should be sufficient unless T_i is excessively large.

On the other hand, if instead of a photon, a second electron carries away the excess energy and momentum, then the process is called *three-body recombination.* The reaction rate density is

$$\Gamma_{3\mathrm{br}} = S_{3\mathrm{br}} n_e^2 n_i \tag{2.47}$$

Since this is the inverse process to collisional ionization, the rate coefficient $S_{3\mathrm{br}}$ can be found by equating the two reaction rate densities at equilibrium:

$$S_{\mathrm{ci}} \left[n_e n_g\right]_{\mathrm{eq}} = S_{3\mathrm{br}} \left[n_e^2 n_i\right]_{\mathrm{eq}} \tag{2.48a}$$

$$S_{3\mathrm{br}} = S_{\mathrm{ci}} \left[\frac{n_g}{n_e n_i}\right]_{\mathrm{eq}}. \tag{2.48b}$$

The bracketed term in Equation 2.48 is just the equilibrium constant, which for a singly-charged ion is given by Saha relation:

$$\left[\frac{n_e n_i}{n_g}\right]_{\mathrm{eq}} = \frac{2g_i}{g_n} \left(\frac{2\pi m_e k_B T_e}{h^2}\right)^{3/2} \exp\frac{-\mathcal{E}_b}{k_B T_e}. \tag{2.49}$$

Also as the opposite reaction of collisional ionization, energy is gained by the electron fluid, and the rate of energy gain is

$$q_{3\mathrm{br}}^e = \Gamma_{3\mathrm{br}} \mathcal{E}_b. \tag{2.50}$$

2.6 Laser interactions

While the model assumes the droplet is at least partially ionized, the plasma state is not formed without a laser source. The laser introduces a fourth particle – the photon – into the system, and it is quickly absorbed by either bound or free electrons. Whereas the term "electrons" has typically referred to free electrons for much of this chapter, in this context it collectively refers to both free and bound electrons.

2.6.1 Energy deposition

This model assumes the laser beam is a line heat source within each droplet even if the laser cross-sectional area is known. The volumetric heat generation, as a result, contains a Dirac delta function $\delta(\boldsymbol{r}_{\mathrm{laser}})$ where $\boldsymbol{r}_{\mathrm{laser}}$ is the laser trajectory. Finite-volume methods blend well with this assumption since the delta function is always integrated out during the formulation.

$$\iiint_{\Gamma \supset \boldsymbol{r}_0} \delta(\boldsymbol{r}_0) dV = 1 \tag{2.51}$$

Therefore, the total energy deposited in a cell that the laser passes through is the difference in laser energy at the cell entrance and exit. If the entrance energy upon cell k is $I_k A$, it will have been attenuated by a factor of $e^{-\alpha_k s_k}$ when exitting the cell – s_k being the distance that the laser beam travels within cell k. Therefore,

$$\iiint_{V_k} q_{\mathrm{laser}}''' dV = I_k A(1 - e^{-\alpha_k s_k}) \tag{2.52}$$

Bound and free electrons differ in how the energy is spent. A free electron, upon absorbing a photon, simply increases in kinetic energy. A bound electron, however, can either be excited or removed from its orbit altogether. If excitation occurs, collisions with the atom causes the electron to rapidly relax and transfer the acquired energy into thermal energy of the bulk liquid; therefore, on the timescale of interest, it is responsible to assume that the energy goes directly into heating the neutral fluid.

With two channels in which laser energy can be deposited, the total attenuation coefficient α is the sum of the bound and free counterparts.

$$\alpha = \alpha_b + \alpha_{\mathrm{ib}}, \tag{2.53}$$

and the fraction of energy that goes into each channel is proportional to the relative magnitude of the two absorption coefficients

$$q_b = I_0 A(1 - e^{-\alpha s})\frac{\alpha_b}{\alpha} \tag{2.54a}$$

$$q_{\gamma e} = I_0 A(1 - e^{-\alpha s})\frac{\alpha_{\mathrm{ib}}}{\alpha}. \tag{2.54b}$$

The subscript "ib" in Equation 2.53 refers to inverse bremsstrahlung, a process in which free electrons, in the vincinty of an ion, absorbs energy and starts to accelerate. The double subscript γe infers as if the heat source were due to a "collision" between a photon and a free electron, similar to how other heat sources are denoted in Section 2.3.1. Inverse bremsstrahlung should take over as the dominant absorption process once a sufficiently opaque medium is formed. The attenuation coefficient will increase proportionally to the inverse bremsstrahlung frequency ν_{ib}

$$\alpha_{\mathrm{ib}} = \frac{\nu_{\mathrm{ib}}\eta}{c} = \frac{\omega_e^2}{\omega^2 + \nu_{ei}^2}\frac{\nu_{ei}\eta}{c}, \tag{2.55}$$

where ω_e is the plasma electron frequency and ω is the laser frequency. The former is defined as

$$\omega_e^2 \equiv \frac{n_e e^2}{\varepsilon_0 m_e}. \tag{2.56}$$

When $\omega \gg \nu_{ei}$, ν_{ib} is usually given in the form

$$\nu_{\mathrm{ib}} = \frac{n_e}{n_c}\nu_{ei}, \tag{2.57}$$

and the critical density n_c is defined in terms of ω

$$\omega^2 = \frac{n_c e^2}{\varepsilon_0 m_e}. \tag{2.58}$$

2.6.2 Photoionization

A bound electron requires a binding energy, $\mathcal{E}_b$, to remove it from the orbit. When a laser beam deposits energy into an atom, each photon contributes an energy of $h\nu$. Therefore, the minimum number of photons needed to completely remove an electron from its orbit is

$$N \equiv \left\lceil \frac{\mathcal{E}_b}{h\nu} \right\rceil. \tag{2.59}$$

The N-photon ionization rate is proportional to the N-th power of a particular measure of photon flux. In Mihelic (2021), that measure is the photon scalar flux, $\phi = \dfrac{I}{h\nu}$ [6]. However, since the laser intensity monotonic decreases with distance travelled, it is better to calculate the ionization rate using a path-averaged scalar flux within a computational cell k:

$$\overline{\phi}_k = \frac{\overline{I}_k}{h\nu} \tag{2.60a}$$

$$= \frac{1}{h\nu s_k}\int_0^{s_k} I_0 e^{-\alpha_k s} ds \tag{2.60b}$$

$$= \frac{I_0}{h\nu}\frac{1-e^{-\alpha_k s_k}}{\alpha_k s_k} \tag{2.60c}$$

The rate is then given by

$$\Gamma_{\text{pi}} = S_{\text{pi}}^{(N)}\overline{\phi}^N n_g \tag{2.61}$$

Here, $S_{\text{pi}}^{(N)}$ is the N-photon ionization rate coefficient and n_g is the neutral particle number density in the gas phase. In order for Γ_{pi} to have the correct unit, the unit of $S_{\text{pi}}^{(N)}$ is $\text{m}^{2N}\ \text{s}^{N-1}$. If $N = 1$, then $S_{\text{pi}}^{(1)}$ has the expected units of cross section or area. For that reason, $S_{\text{pi}}^{(N)}$ is often referred to as a cross section in literature – regardless of N and how photon concentration is expressed (other than $\overline{\phi}$) – despite the fact that it can assume different units. For consistency, the symbol σ reserved for cross section is only used to denote quantities with units of area.

In the case of single-photon ionization, $S_{\text{pi}}^{(1)}$ is related to the absorption coefficient α_b by

$$S_{\text{pi}}^{(1)} = \Phi\frac{\alpha_b}{n_g}, \tag{2.62}$$

where $\Phi \in [0,1]$ is the quantum efficiency, which denotes the fraction of absorbed photons that actually induce ionization. If $\Phi = 1$, then α_b is equivalent to a macroscopic cross section.

Upon ionization, the electron fluid gains an amount of energy equivalent to

$$q_{\text{pi}}^e = (Nh\nu - \mathcal{E}_b)\Gamma_{\text{pi}}, \tag{2.63}$$

while any energy from the initial laser beam not spent on ionizing will instead serve as a heat source for the neutral atoms:

$$q_{\gamma n}''' = q_b''' - Nh\nu\Gamma_{\text{pi}}. \tag{2.64}$$

For simplicity, bound electrons are assumed to only absorb N photons prior to removal from its parent nucleus. Above-threshold ionization, a phenomenon where an electron absorbs more than N photons, is also possible, but will be neglected here.

2.7 Summary

The system of equations have a total of 27 scalar variables, organized into five categories. The liquid phase equations are

$$\frac{\partial \alpha_l}{\partial t} + \nabla \cdot \alpha_l \mathbf{u}_l = \mathcal{P}_{gl} \tag{2.65}$$

$$\frac{\partial \alpha_l \rho_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l = -\Sigma_{\text{evap}} \tag{2.66}$$

$$\frac{\partial \alpha_l \rho_l \mathbf{u}_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l \otimes \mathbf{u}_l = -\nabla p_l + \nabla \cdot \mathbf{T}_l + \rho_l \mathbf{g} + \mathbf{f}_{gl} \tag{2.67}$$

$$\frac{\partial \alpha_l \rho_l E_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l H_l = -\nabla \cdot \mathbf{q}_l'' + \nabla \cdot [\mathbf{T}_l \cdot \mathbf{u}_l] + \sum_j^{g,\gamma} q_{jl}''' \tag{2.68}$$

The gas phase equations are

$$\frac{\partial \alpha_g \rho_g}{\partial t} + \nabla \cdot \rho_g \mathbf{u}_g = \frac{M}{N_A} \sum_r \Gamma_r^g \tag{2.69}$$

$$\frac{\partial \alpha_g \rho_g \mathbf{u}_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g \otimes \mathbf{u}_g = -\nabla p_g + \nabla \cdot \mathbf{T}_g + \rho_g \mathbf{g} + \sum_j^{l,e,i} \mathbf{f}_{jg} + \frac{M}{N_A} \sum_r \Gamma_r^g \mathbf{u}_r \tag{2.70}$$

$$\frac{\partial \alpha_g \rho_g E_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g H_g = -\nabla \cdot \mathbf{q}_g'' + \nabla \cdot [\mathbf{T}_g \cdot \mathbf{u}_g] + \sum_j^{l,e,i,\gamma} q_{jg}''' + \frac{M}{N_A} \sum_r \Gamma_r^g E_r \tag{2.71}$$

The electron equations are

$$\frac{\partial \alpha_g n_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e = \sum_r \Gamma_r^e \tag{2.72}$$

$$m_e \left(\frac{\partial \alpha_g n_e \mathbf{u}_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e \otimes \mathbf{u}_e \right) = -\nabla p_e - e \alpha_g n_e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B}) + \sum_j^{g,i} \mathbf{f}_{je} \tag{2.73}$$

$$\frac{\partial \alpha_g n_e \mathcal{E}_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e \mathcal{H}_e = -e \alpha_g n_e \mathbf{u}_e \cdot \mathbf{E} + \sum_j^{g,i,\gamma} q_{je}''' + \sum_r \Gamma_r^e (\mathcal{E}_r - \mathcal{E}_b) \tag{2.74}$$

The ion equations are

$$\frac{\partial \alpha_g n_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i = \sum_r \Gamma_r^i \tag{2.75}$$

$$m_i \left(\frac{\partial \alpha_g n_i \mathbf{u}_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i \otimes \mathbf{u}_i \right) = -\nabla p_i + e \alpha_g n_i (\mathbf{E} + \mathbf{u}_i \times \mathbf{B}) + \sum_j^{g,e} \mathbf{f}_{ji} + m_i \sum_r \Gamma_r^i \mathbf{u}_r \tag{2.76}$$

$$\frac{\partial \alpha_g n_i \mathcal{E}_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i \mathcal{H}_i = e \alpha_g n_i \mathbf{u}_i \cdot \mathbf{E} + \sum_j^{g,e} q_{ji}''' + \sum_r \Gamma_r^i \mathcal{E}_r \tag{2.77}$$

Finally, Maxwell's Equations are needed to model the electric and magnetic fields.

$$\varepsilon_r \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\mu_0} \nabla \times \mathbf{B} - e\alpha_g (n_i \mathbf{u}_i - n_e \mathbf{u}_e) \tag{2.78}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} \tag{2.79}$$

Chapter 3

Numerical methods

Bibliography

[1] S. I. Braginskii. "Transport Processes in a Plasma". In: *Reviews of Plasma Physics* (1965).

[2] Francis F. Chen. *Introduction to Plasma Physics and Controlled Fusion*. 3rd ed. Springer, 2016.

[3] Flash Center for Computational Science. *FLASH User's Guide*. Version 4.8. 2024.

[4] E. M. Epperlein and M. G. Haines. "Plasma transport coefficients in a magnetic field by direct numerical solution of the Fokker–Planck equation". In: *Physics of Fluids* 29.1029–1041 (4 1986). DOI: `10.1063/1.865901`.

[5] Richard Fitzpatrick. *CRC Press, Taylor & Francis Group*. Plasma Physics: An Introduction, 2014. ISBN: 978-1-4665-9426-5.

[6] Andrej Mihelic and Martin Horvat. "Calculation of multiphoton ionization amplitudes and cross sections of few-electron atoms". In: *Phys. Rev. A* 103 (4 Apr. 2021), p. 043108. DOI: `10.1103/PhysRevA.103.043108`. URL: `https://link.aps.org/doi/10.1103/PhysRevA.103.043108`.

[7] Philip M. Stone and Yong-Ki Kim. "An overview of the BEB method for electron-impact ionization of atoms and molecules". In: *Surface and Interface Analysis* 37 (2005). DOI: `10.1002/sia.2089`.